

Module Specification

Module Title: Applied Research & Engineering Practice I	Module Code: EG7030 Level: 7 Credit: 60 ECTS credit: 30	Module Leader: Dr Ali Abbas
Pre-requisite: None		Pre-cursor: None
Co-requisite: None	Excluded combinations : None	Suitable for incoming study abroad? N
Location of delivery: UEL		
Summary of module for applicants: Applied Research skills are key to addressing engineering challenges and providing creative solutions. The module aims to provide students with an opportunity to explore a research topic in depth, with appropriate research methods, Interpret the results in an engineering context, provide critical discussion and conclusions, and compose a research dissertation. The students will undertake this in subject areas relevant to the programme they are enrolled on. It also allows them to plan and execute their work plan and manage their time as well as exercising their engineering judgment and applying principles of engineering practice. They are encouraged to experience engineering as problem solving, considering the systematic processes that engineers use to define and resolve problems.		
Main topics of study: • Students will select a research topic in consultation with an appropriate subject specialist supervisor. • Students are expected to develop a project proposal with a time management plan will be developed in liaison with the project supervisor. • Students are expected to use and integrate the knowledge and skills gained through the taught modules in MSc programmes (e.g. Civil, Structural Engineering). • Students are expected to develop communication and research methodology skills through Research Methods classes. • Students are encouraged to expand their knowledge and skills on engineering ethics and practice and to enhance key skills such as problem solving and synthesis of information to make valid judgements and reasoning.		
This module will be able to demonstrate at least one of the following examples/ exposures <i>(please tick one or more of the appropriate boxes, evidence will need to be provided later in this document)</i> Live, applied project ☒ Company/engagement visits ☐ Company/industry sector endorsement/badging/sponsorship/award ☒		

Learning Outcomes for the module

Code of Competencies

- *Digital Proficiency - Code = (DP)*
- *Industry Connections - Code = (IC)*
- *Emotional Intelligence Development - Code = (EID)*
- *Social Intelligence Development - Code = (SID)*
- *Physical Intelligence Development - Code = (PID)*
- *Cultural Intelligence Development - Code = (CID)*
- *Community Connections - Code = (CC)*
- *UEL Give-Back - Code = (UGB)*

At the end of this module, students will be able to:

Knowledge

1. Identify and apply appropriate theoretical framework to the chosen topic.
2. Demonstrate detailed knowledge of one chosen and highly specific area within the domain of their programme of study and communicate this knowledge through a written report.
3. Demonstrate knowledge of research methods appropriate for Masters level (i.e. Level7) research and to communicate this knowledge through a written report.

Thinking skills

4. Implement a piece of advanced research as formulated in the project proposal.
5. Competently and critically select appropriate research methods and tools for data collection and analysis. (DP)
6. Critically interpret the results in relation to existing knowledge.
7. Demonstrate critical self-reflection on the research process and suggest further developments.

Subject-based practical skills

8. Conduct an advanced search of literature and/or other appropriate sources.

Skills for life and work (general skills)

9. Prepare a detailed piece of work written in a scholarly fashion according to set guidelines.
10. Demonstrate the confidence and skills to manage research in a way that is consistent with both professional practice and the normal principles of research ethics. (IC)
11. Plan the production of an extended piece of work to meet a fixed delivery deadline. (EID)

Teaching/ learning methods/strategies used to enable the achievement of learning outcomes:

For on campus students:

Lectures to develop research method and communication skills.
Online continuous learning.
Independent study complimented by supervisor – student meetings. (SID)

Assessment methods which enable students to demonstrate the learning outcomes for the module; please define as necessary: Written Submission (12,500 words)	Weighting: 100%	Learning Outcomes demonstrated: 1-11
Reading and resources for the module: These must be up to date and presented in correct Harvard format unless a Professional Body specifically requires a different format Core UEL Moodle: Academic Integrity - Resources for Students→ (in particular: Academic Integrity Policy, Cite Them Right, Turnitin resources). Bell, J. (2010) <i>Doing your research project</i>. 5th edn. Buckingham: Open University Press. Hart, C. (1998) <i>Doing a literature review: Releasing the social science research imagination</i>. London: Sage Publications. O’Leary, Z. (2017) <i>The essential guide to doing your research project</i>. 3rd edn. Sage Publications Ltd. Swetnam, D. (2000) <i>Writing your dissertation</i>. 3rd edn. Oxford: How to Books. Pears, R. and Shields, G. (2013) <i>Cite them right: the essential referencing guide</i>. 9th edn. Basingstoke: Palgrave Macmillan. Recommended Rudestam, K.E. and Newton, R.R. (2007) <i>Surviving your dissertation: a comprehensive guide to content and process</i>. London: Sage. Sharp, J.A. and Howard, K. (2002) <i>The management of a student research project</i>. 3rd edn. Farnham: Gower Publishing Ltd. Glasman-Deal, H. (2009) <i>Science research writing for non-native speakers of English</i>. London: Imperial College Press Levin, P. (2008) <i>Excellent dissertation</i>. Maidenhead: Open University Press Walliman, N. S. (2011) <i>Your research project: a step-by-step guide for the first-time researcher</i>. 3rd edn. London: Sage. Farrell, P (2011) <i>Writing a built environment dissertation. Practical guidance and examples</i>. Chichester: John Wiley & Sons Ltd.		
Provide evidence of how this module will be able to demonstrate at least one of the following examples/ exposures <i>Live, applied project</i> Data extracted for analysis and written research report can be from live projects in engineering field (e.g. civil, mechanical etc.) <i>Company/engagement visits</i> N/A <i>Company/industry sector endorsement/badging/sponsorship/award</i> Invite guest speakers from industry to cover specialist areas in engineering such as health and safety, engineering ethics and professional practice, mental wealth etc. (IC)		

Indicative learning and teaching time (10 hrs per credit):	Activity
1. Student/tutor interaction: 18 hours 10 hours	Research Methods classes Supervisory meetings
2. Student learning time: 572 hours	
Total hours (1 and 2):	600 hours

